

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Modular Unit 1

Adaptive Classified

OL-T44 · Circles, Arcs and Sectors

With Answers

Questions with full worked solutions

■ 46 pages

FAMILY MAP

Circles Arcs Sectors

F01	Use arc and sector measures	5 Q 14 marks
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F02	Resolve compound and shaded circle regions	14 Q 66 marks
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F03	Use full-circle and semicircle measures	1 Q 5 marks
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Working Unit 08

4MA1-1H Q11

ORIGINAL QUESTION**2 marks**

- 11 $OABC$ is a sector of a circle, centre O

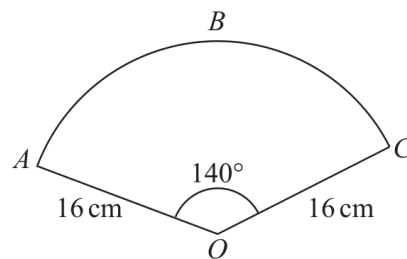


Diagram **NOT**
accurately drawn

$$\text{angle } AOC = 140^\circ$$

$$OA = OC = 16\text{ cm}$$

Calculate the area of the sector.

Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 11 is 2 marks)

Worked Solution 08

Area of a sector

UNIT 1
2 marks**Use the sector fraction**

$$A = \frac{140}{360} \pi (16)^2 = \frac{896\pi}{9} = 312.763 \dots$$

Final answer

$$313 \text{ cm}^2 \text{ (3 s.f.)}$$

Question 14

4MA1-1H Q20

ORIGINAL QUESTION

4 marks

- 20 The diagram shows a sector OAC of a circle centre O

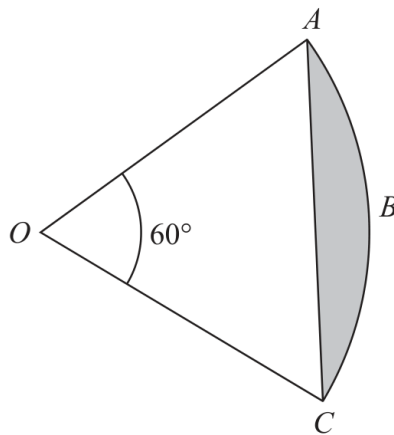


Diagram **NOT**
accurately drawn

Angle $AOC = 60^\circ$

The area of the shaded segment ABC is 38 cm^2

Work out the perimeter of the shaded segment ABC
Give your answer correct to one decimal place.

..... cm

Worked Solution 14

Circular segment

MODULAR UNIT 1

4 marks

Radius

$$38 = r^2(\pi/6 - \sqrt{3}/4) \Rightarrow r = 20.4815 \dots$$

Perimeter

$$r + \pi r/3 = 41.9 \text{ cm.}$$

Final answer

41.9 cm

Question 08

4MA1-1H Q5 demand

ORIGINAL QUESTION UNIT 1

3 marks

- 5 The diagram shows a shaded shape made from three identical semicircles, *X*, *Y* and *Z*

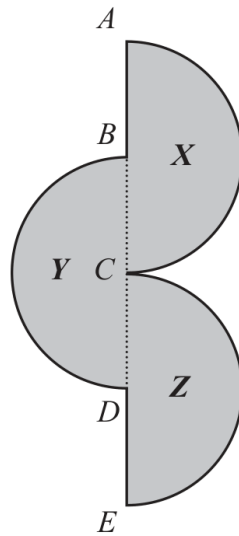


Diagram **NOT**
accurately drawn

ABCDE is a straight line.

AC is the diameter of semicircle *X* and *B* is the centre of semicircle *X*

BD is the diameter of semicircle *Y* and *C* is the centre of semicircle *Y*

CE is the diameter of semicircle *Z* and *D* is the centre of semicircle *Z*

$$AC = BD = CE = 20 \text{ cm}$$

Work out the perimeter of the shaded shape.

Give your answer correct to the nearest whole number.

..... cm

Worked solution 08

Perimeter of three semicircles

MODULAR UNIT 1

3 marks

Demand – Perimeter of three semicircles

Add curved parts

$$3 \left(\frac{1}{2} \pi \times 20 \right) = 30\pi.$$

Add exposed diameters

$$AB + DE = 10 + 10 = 20; \quad P = 30\pi + 20 \approx 114.25.$$

Final answer

114 cm

Question 30

4MA1-2H Q20 whole

ORIGINAL QUESTION UNIT 1

4 marks

- 20 The diagram shows equilateral triangle ABC with sides of length 10 cm. A circle is drawn inside the triangle.

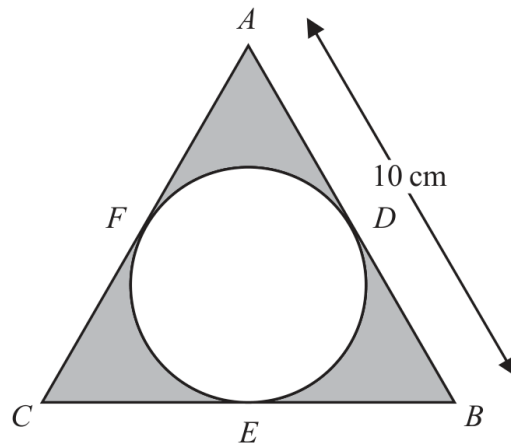


Diagram **NOT**
accurately drawn

D , E and F are points on the circle.

ADB , BEC and CFA are tangents to the circle.

Calculate the total area of the regions shown shaded in the diagram.
Give your answer correct to 3 significant figures.

..... cm^2

Worked solution

METHOD / WORKING

MODULAR UNIT 1 • NOVEMBER 2023 DETAILED SOLUTIONS
Worked solution 30 MODULAR UNIT 1
Shaded area between an equilateral triangle and its incircle 4 marks
whole – Shaded area between an equilateral triangle and its incircle
Find the triangle area
$A_{\triangle} = 1$
$2(10)(10) \sin 60^\circ = 25$
$\sqrt{}$
3.
Find the inradius
$r = 10$
$\sqrt{}$
3
$6 = 5$
$\sqrt{}$
3
3 .
Subtract the circle area
Ashaded = 25
$\sqrt{}$
$3 - \pi r^2 = 25$
$\sqrt{}$
$3 - 25\pi$
3 .
Round
Ashaded = 17.121 . . . $\approx 17.1 \text{ cm}^2$.
Final answer
17.1 cm^2

FINAL ANSWER

Question 11

4MA1-1H Q22

ORIGINAL QUESTION

6 marks

- 22 The diagram shows two circles with centre O and a regular pentagon $ABCDE$

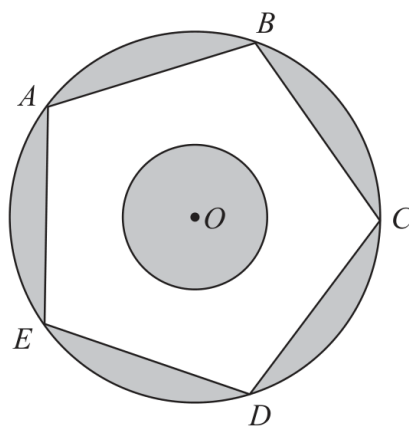


Diagram **NOT**
accurately drawn

A , B , C , D and E are points on the larger circle.
The pentagon has sides of length 8 cm.

The diagram is shaded such that

$$\text{shaded area} = \text{unshaded area}$$

Work out the radius of the smaller circle.
Give your answer correct to 3 significant figures.

..... cm

Worked Solution 11

Equal shaded and unshaded areas

MODULAR UNIT 1**6 marks****Find the circumradius**

$$R = \frac{4}{\sin 36^\circ} = 6.805 \dots$$

Find the two main areas

$$A_{\text{circle}} = \pi R^2 = 145.489 \dots, \quad A_{\text{pentagon}} = 5 \left(\frac{1}{2} R^2 \sin 72^\circ \right) = 110.111 \dots$$

Use the equal-area condition

$$A_{\text{circle}} - A_{\text{pentagon}} + \pi r^2 = A_{\text{pentagon}} - \pi r^2.$$

Solve

$$r = \sqrt{\frac{2A_{\text{pentagon}} - A_{\text{circle}}}{2\pi}} = 3.44 \dots$$

Final answer

3.45 cm

Original question 12

4MA1-1H Q16 M01

MODULAR UNIT 1

4 marks

16 $OAPB$ is a sector of a circle, centre O

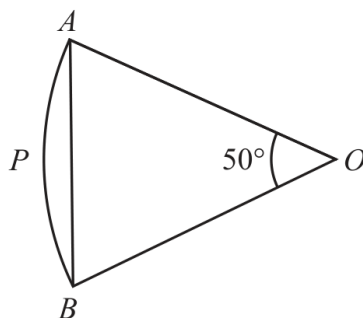


Diagram **NOT**
accurately drawn

Angle $AOB = 50^\circ$

Area of triangle $OAB = 120 \text{ cm}^2$

Work out the area of the sector $OAPB$

Give your answer correct to 3 significant figures.

..... cm^2

Worked solution 12

Find the area of a sector

MODULAR UNIT 1

4 marks

M01 – Find the area of a sector**Use the triangle area**

$$120 = \frac{1}{2}r^2 \sin 50^\circ \Rightarrow r^2 = 240 / \sin 50^\circ.$$

Find r

$$r = 17.7002 \dots$$

Use the sector fraction

$$A = \frac{50}{360} \pi r^2 = 136.7 \dots$$

Round

$$A = 137 \text{ cm}^2 \text{ (3 s.f.)}.$$

Final answer

$$137 \text{ cm}^2$$

Original question 28

4MA1-2H Q25 M01

MODULAR UNIT 1

5 marks

- 25 The diagram shows an equilateral triangle ABC and a circle with centre O

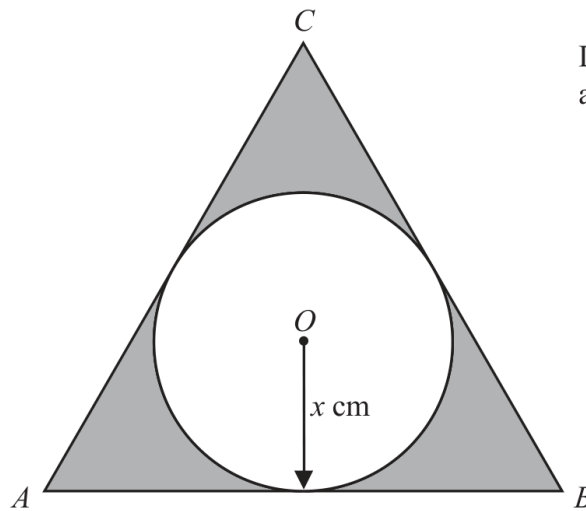


Diagram **NOT**
accurately drawn

AB , BC and CA are tangents to the circle.

The radius of the circle is x cm

The total area, in cm^2 , of the regions shown shaded in the diagram is nx^2

Find the value of n

Give your answer correct to 3 significant figures.

Original question 28 (continued)

4MA1-2H Q25 M01

MODULAR UNIT 1

5 marks

 $n = \dots\dots\dots$

Worked solution 28

Area of shaded regions in an equilateral triangle

MODULAR UNIT 1

5 marks

M01 – Area of shaded regions in an equilateral triangle**Find the side length**The inradius is x , so the side is $2\sqrt{3}x$.**Find triangle area**

$$A_{\text{triangle}} = \frac{1}{2}(2\sqrt{3}x)(3x) = 3\sqrt{3}x^2.$$

Subtract the circle

$$A_{\text{shaded}} = (3\sqrt{3} - \pi)x^2.$$

Read n

$$n = 3\sqrt{3} - \pi = 2.0548 \dots$$

Round

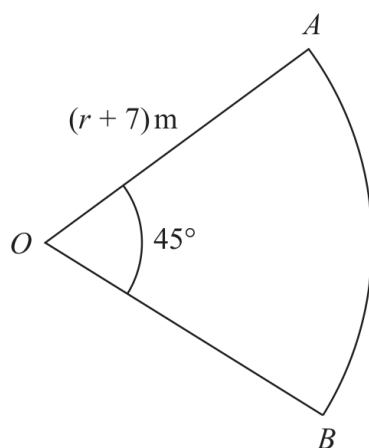
$$n = 2.05 \text{ (3 s.f.)}.$$

Final answer

2.05

Original question 13

4MA1-1H Q19 Q19

ORIGINAL QUESTION**5 marks****19**Diagram **NOT**
accurately drawn

OAB is a sector S of a circle with centre O and radius $(r + 7)$ metres.
Angle $AOB = 45^\circ$

A circle C has radius $(r - 2)$ metres.

The area of sector S is twice the area of circle C

Find the value of r

Show your working clearly.

Question 19 continued. $r = \dots\dots\dots$

Worked solution

Family: Resolve compound and shaded circle regions • Variant: Solve an algebraic circle-measure con-

METHOD / WORKING

MODULAR UNIT 1 • JUNE 2022 DETAILED SOLUTIONS
Worked solution 13 MODULAR UNIT 1
Chapter: Circles, Arcs & Sectors 5 marks
Q19 – Chapter: Circles, Arcs & Sectors
Family: Resolve compound and shaded circle regions • Variant: Solve an algebraic circle-measure con-
straint
Method
The sector area is $(45/360)\pi(r + 7)^2$.
Why: A 45-degree sector is 45/360 of its full circle.
Method
The condition gives $(45/360)\pi(r + 7)^2 = 2\pi(r - 2)^2$.
Why: The sector area is twice the area of circle C.
Method
<i>After cancelling π and simplifying, $(r + 7)^2 = 16(r - 2)^2$, equivalently $5r^2 - 26r + 5 = 0$.</i>
Why: Expand or rearrange the exact area equation.
Method
$5r^2 - 26r + 5 = (5r - 1)(r - 5)$, so $r = 1/5$ or $r = 5$.
Why: Factorise and apply the zero-product rule.
Method
A circle of radius $r - 2$ requires $r > 2$, so $r = 1/5$ is invalid and $r = 5$.
Why: Use the physical positivity of the stated radius.
Final answer
$r: r = 5$

FINAL ANSWER

Original question 27

4MA1-2H Q20 Q20

ORIGINAL QUESTION**4 marks**

- 20 The diagram shows four identical circles drawn inside a square.

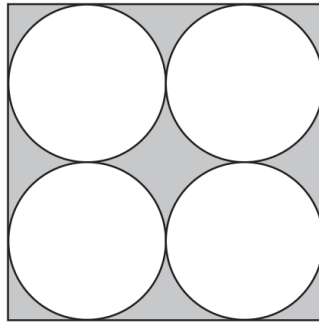


Diagram **NOT**
accurately drawn

Each circle touches two other circles and two sides of the square.

The region inside the square that is outside the circles, shown shaded in the diagram, has a total area of 40 cm^2

Work out the perimeter of the square.

Give your answer correct to 3 significant figures.

..... cm

Worked solution

Family: Resolve compound and shaded circle regions • Variant: Find annular, overlapping or multi-circle

METHOD / WORKING

MODULAR UNIT 1 • JUNE 2022 DETAILED SOLUTIONS
Worked solution 27 MODULAR UNIT 1
Chapter: Circles, Arcs & Sectors 4 marks
Q20 – Chapter: Circles, Arcs & Sectors
Family: Resolve compound and shaded circle regions • Variant: Find annular, overlapping or multi-circle
area
Method
Let each circle have radius r ; the square side is $4r$.
Why: Two diameters fit along each side.
Method
The shaded area is $(4r)^2 - 4\pi r^2 = 40$, so $r^2 = 10/(4 - \pi)$.
Why: Subtract the four circle areas from the square area.
Method
The square perimeter is $16r = 16 \sqrt{10/(4 - \pi)}$.
Why: Perimeter is four times the side $4r$.
Method
This is 54.608... cm, so the perimeter is 54.6 cm to three significant figures.
Why: Round the final value once.
Final answer
perimeter: 54.6 cm

FINAL ANSWER

Original question 14

4MA1-1H Q16

ORIGINAL QUESTION**4 marks**

16 The diagram shows a circle with centre O

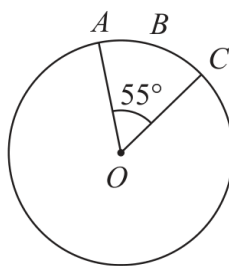


Diagram **NOT**
accurately drawn

A , B and C are points on the circle so that the length of the arc ABC is 5 cm.

Given that angle $AOC = 55^\circ$

work out the area of the circle.

Give your answer correct to one decimal place.

..... cm^2

Worked solution

Family: Use arc and sector measures • Variant: Recover radius or central angle from an arc/sector measure

METHOD / WORKING

MODULAR UNIT 1 • JANUARY 2022 DETAILED SOLUTIONS
Worked solution 14 MODULAR UNIT 1
Chapter: Circles, Arcs & Sectors 4 marks
Q16 – Chapter: Circles, Arcs & Sectors
Family: Use arc and sector measures • Variant: Recover radius or central angle from an arc/sector measure
Method
$5 = (55/360) \times 2\pi r$, so $r = 5 \times 360 / (55 \times 2\pi)$.
Why: Use the arc-length formula $s = (\theta/360)2\pi r$.
Method
$r \approx 5.207$ cm.
Why: Evaluate the radius.
Method
$Area = \pi r^2 \approx 85.2 \text{ cm}^2$.
Why: Use $\pi \approx 3.14$.
Method
To 1 decimal place, the area is 85.2 cm ² .
Why: Round as requested.
Final answer
area : 85.2 cm^2

FINAL ANSWER

- 11 The diagram shows a shaded shape $ABCD$ made from a semicircle ABC and a right-angled triangle ACD .

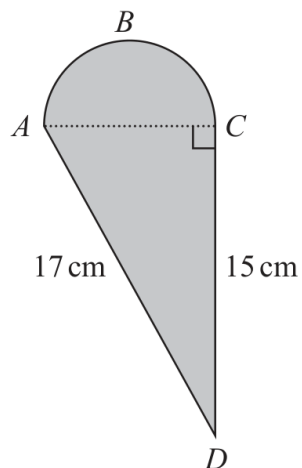


Diagram **NOT**
accurately drawn

AC is the diameter of the semicircle ABC .

Work out the perimeter of the shaded shape.

Give your answer correct to 3 significant figures.

..... cm

Worked solution

January 2020 | Question 11 | 5 marks

Family: Resolve compound and shaded circle regions | Variant: Find a mixed straight-and-curved perimeter

METHOD / WORKING

$$1. AC^2 = 17^2 - 15^2 = 64, \text{ so } AC = 8\text{ cm.}$$

$$2. \text{ Semicircle radius} = 4 \text{ cm.}$$

$$3. \text{ Semicircle arc length} = \pi r = 4\pi.$$

$$4. \text{ Perimeter} = 17 + 15 + 4\pi = 44.566\dots \text{ cm.}$$

$$5. \text{ To 3 s.f., } 44.6 \text{ cm.}$$

FINAL ANSWER

44.6 cm

- 8 The region, shown shaded in the diagram, is a path.

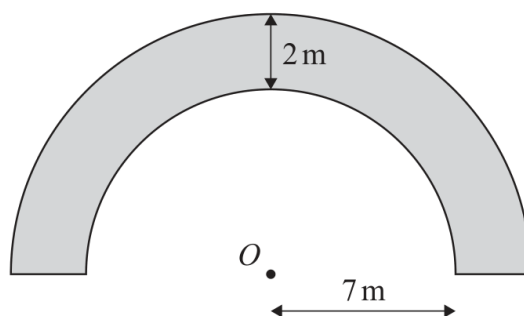


Diagram **NOT**
accurately drawn

The boundary of the path is formed by two semicircles, with the same centre O , and two straight lines.

The inner semicircle has a radius of 7 metres.

The path has a width of 2 metres.

Work out the perimeter of the path.

Give your answer correct to one decimal place.

..... m

Worked solution

January 2021 | Question 8 | 3 marks

Family: Resolve compound and shaded circle regions • Variant: Find a mixed straight-and-curved perimeter

Method / working

1. The inner semicircle arc is 7π and the outer arc is 9π .
 2. Add the two straight widths: perimeter = $7\pi + 9\pi + 4 = 16\pi + 4$.
 3. $16\pi + 4 = 54.265\dots \sim 54.3$ m.
- Final answer: 54.3 m

- 8 A , B and C are points on a circle with centre O .

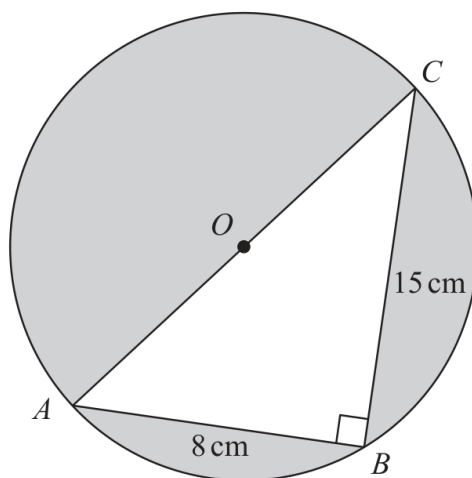


Diagram **NOT**
accurately drawn

AOC is a diameter of the circle.

$AB = 8 \text{ cm}$ $BC = 15 \text{ cm}$

Angle $ABC = 90^\circ$

Work out the total area of the regions shown shaded in the diagram.
Give your answer correct to 3 significant figures.

..... cm^2

Worked solution

January 2021 | Question 8 | 5 marks

Family: Resolve compound and shaded circle regions | Variant: Find annular, overlapping or multi-circle area

METHOD / WORKING

1. AC is the diameter: $AC = \sqrt{8^2 + 15^2} = 17$ cm.2. Circle area = $\pi(17/2)^2$.3. $\text{Triangle ABC area} = \frac{1}{2} \times 8 \times 15 = 60 \text{ cm}^2$.

4. Shaded area = circle area - triangle area.

5. This gives 166.99... 167 cm^2 .**FINAL ANSWER**167 cm^2

22 The diagram shows a sector OBC of a circle with centre O and radius $(6 + x)$ cm.

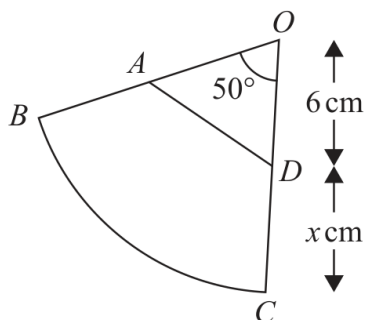


Diagram **NOT**
accurately drawn

A is the point on OB and D is the point on OC such that $OA = OD = 6$ cm

Angle $BOC = 50^\circ$

Given that

the perimeter of sector $OBC = 2 \times$ the perimeter of triangle OAD

find the value of x .

Give your answer correct to 3 significant figures.

$x = \dots\dots\dots$

Worked solution

January 2021 | Question 22 | 6 marks

Family: Resolve compound and shaded circle regions • Variant: Solve an algebraic circle-measure constraint

Method / working

1. Use cosine rule in triangle OAD to find $AD = 2 \times 6 \sin 25^\circ \sim 5.07$.
2. Triangle perimeter $= 6 + 6 + AD \sim 17.1$.
3. Sector arc $= \frac{50}{360} \times 2\pi(6+x)$.
4. Set sector perimeter equal to twice the triangle perimeter.
5. Solve the resulting linear equation in x .
6. To 3 s.f., $x = 5.89$.

Final answer: $x = 5.89$

12 The diagram shows a sector of a circle with radius 7 cm.

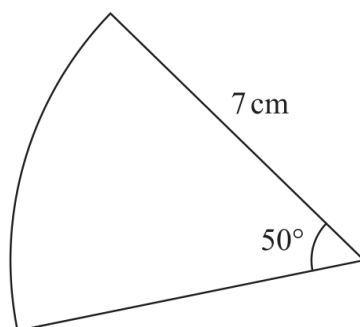


Diagram **NOT**
accurately drawn

Work out the length of the arc of the sector.
Give your answer correct to one decimal place.

..... cm

Worked solution

June 2021 | Question 12 | 2 marks

Family: Use arc and sector measures • Variant: Find arc length

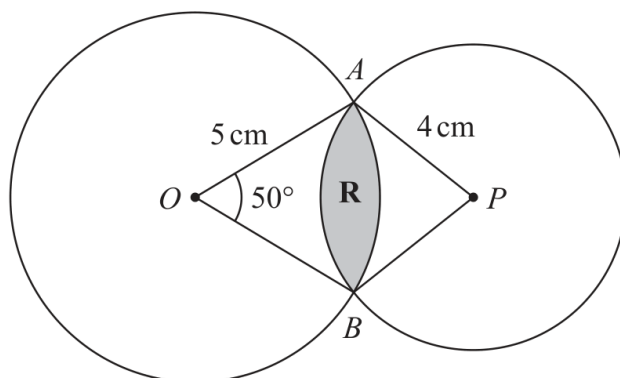
Method / working

1. Arc length = $(50/360) \times 2\pi \times 7$.

2. Evaluate to 6.108... and round appropriately: 6.1 cm.

Final answer: 6.1 cm

- 25 The diagram shows two circles such that the region **R**, shown shaded in the diagram, is the region common to both circles.



One of the circles has centre O and radius 5 cm .
 The other circle has centre P and radius 4 cm .
 Angle $AOB = 50^\circ$

Calculate the area of region **R**.
 Give your answer correct to 3 significant figures.

Diagram **NOT**
 accurately drawn

..... cm^2

Worked solution

June 2021 | Question 25 | 6 marks

Family: Resolve compound and shaded circle regions | Variant: Find annular, overlapping or multi-circle area

METHOD / WORKING

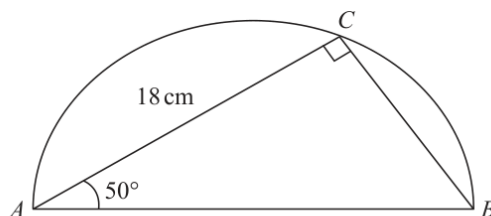
1. Find the common chord AB: $AB = 2 \times 5 \times \sin 25^\circ = 4.226 \dots \text{ cm}$.
2. In the radius-4 triangle, use the cosine rule to find angle APB $\sim 63.77^\circ$.
3. Compute sector AOB $= \frac{50}{360} \pi \cdot 5^2 \sim 10.9$ and triangle OAB $= \frac{1}{2} \cdot 5 \cdot 5 \cdot \sin 50^\circ \sim 9.57$.
4. Compute sector APB $= \frac{63.77}{360} \pi \cdot 4^2 \sim 8.90$ and triangle APB $= \frac{1}{2} \cdot 4 \cdot 4 \cdot \sin 63.77^\circ \sim 7.17$.
5. The overlap is (sector AOB - triangle OAB) + (sector APB - triangle APB).
6. Evaluate and round: 3.06 cm^2 (accepted 3.0 – 3.1).

FINAL ANSWER

 3.06 cm^2

- 8 The diagram shows a triangle ABC inside a semicircle.

Diagram **NOT**
accurately drawn



A , B and C are points on the semicircle.

AB is the diameter of the semicircle.

Angle $ACB = 90^\circ$

Angle $BAC = 50^\circ$

$AC = 18$ cm

Work out the perimeter of the semicircle.

Give your answer correct to 2 significant figures.



..... cm

(Total for Question 8 is 5 marks)

Answer reference | January 2023 | Question 8

Family: Use full-circle and semicircle measures • Variant: Find semicircle area

8	$\cos 50 = \frac{18}{(AB)}$ or $\sin 40 = \frac{18}{(AB)}$ or $\frac{(AB)}{\sin 90} = \frac{18}{\sin 40}$		5	M1	M2 for $(AB =) \sqrt{18^2 + (18 \tan 50)^2}$ oe $(= 28.0030\dots)$ or 28
	$(AB =) \frac{18}{\cos 50} (= 28.0030\dots)$ oe or 28 or $(AB =) \frac{18}{\sin 40} (= 28.0030\dots)$ oe or 28			M1	
	$\frac{1}{2} \times \pi \times "28.0030\dots" (= 43.9\dots)$ oe or 44 $\pi \times "28.0030\dots" (= 87.9\dots)$ oe or 88				M1 for use of πd or $\frac{1}{2} \pi d$ oe Allow any value of $AB > 18$ if M2 not scored
	$"28\dots" + "43.9\dots" (= 71.9900\dots)$ or $"28" + "44"$				M1 ft from previous M1 Allow <i>their d</i> + <i>their</i> $\frac{1}{2} \pi d$
	Correct answer scores full marks (unless from obvious incorrect working)	72			A1 awrt 72
					Total 5 marks

Worked solution

January 2023 | Question 8 | 5 marks

Family: Use full-circle and semicircle measures • Variant: Find semicircle area

Official final mark-scheme pages are embedded immediately after this question.

- 21 The diagram shows the cross section of a circular water pipe.

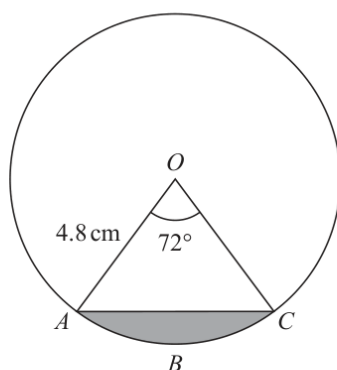


Diagram **NOT**
accurately drawn

OAC is a sector of the circle, centre O

The shaded region in the diagram represents the water flowing in the pipe.

The water flows at 14 cm/s in the pipe.

Work out the volume of water that has flowed through the pipe in 3 minutes.
Give your answer in cm^3 correct to 3 significant figures.



..... cm^3

(Total for Question 21 is 5 marks)

Answer reference | January 2023 | Question 21

Family: Resolve compound and shaded circle regions • Variant: Find a circular segment area or perimeter

21	$\pi \times 4.8^2 \times \frac{72}{360}$ (= 14.4(76...)) oe		5	M1 for finding the area of the sector
	$\frac{1}{2} \times 4.8^2 \times \sin 72$ (= 10.9(56...) or 11) oe or $\frac{1}{2} \times 5.6(4...) \times 3.8(8...) \text{ oe}$			M1 for finding the area of the triangle (Allow use of cosine rule/sine rule/SOHCAHTOA/Pythagoras to find AC (5.6(427.8...)) and OM (3.8(8328...)) where M is the midpoint of AC)
	"14.4(76...)" – "10.9(56...)" (= 3.520...)			M1 for finding the shaded area with all figures from correct working
	"3.5(20...)" $\times 14 \times 3 \times 60$ "3.5(20...)" $\times 2520$			M1
	<i>Award marks within the range from correct working</i>	8870		A1 accept 8820 – 8950 from correct working
				Total 5 marks

Worked solution

January 2023 | Question 21 | 5 marks

Family: Resolve compound and shaded circle regions • Variant: Find a circular segment area or perimeter

Official final mark-scheme pages are embedded immediately after this question.

15 AOB is a sector of a circle with centre O and radius 12 cm

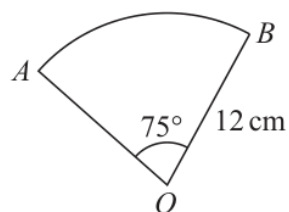


Diagram **NOT**
accurately drawn

Work out the area of the sector.
Give your answer correct to 3 significant figures.

.....cm²

(Total for Question 15 is 2 marks)

Worked solution

November 2025 | Question 15 | 2 marks

Family: Use arc and sector measures | Variant: Recover radius or central angle from an arc/sector measure

METHOD / WORKING

Worked solution
November 2025 Question 15 2 marks
Family: Use arc and sector measures Variant: Recover radius or central angle from an arc/sector measure
Solution
$\text{area} = \frac{75}{360} \times \pi (12)^2$
$= 94.247 \dots$
Answer: 94.2 cm^2 .
PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

FINAL ANSWER

- 6 A large circle with centre O contains 3 identical small circles. Each circle touches two other circles.

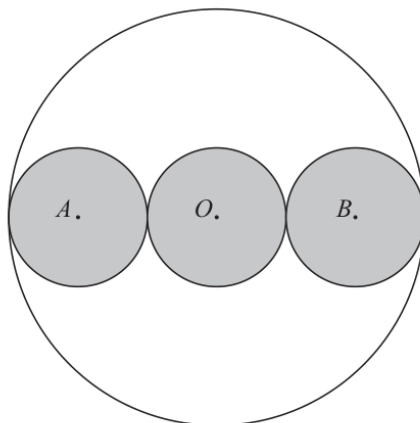


Diagram **NOT**
accurately drawn

A , O and B are the centres of the small circles.
 AOB is a straight line.

The circumference of the large circle is 160 cm

Work out the area of one small circle.
Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 6 is 5 marks)

Worked solution

November 2025 | Question 6 | 5 marks

Family: Resolve compound and shaded circle regions | Variant: Find a mixed straight-and-curved perimeter

METHOD / WORKING

Worked solution
November 2025 Question 6 5 marks
Family: Resolve compound and shaded circle regions Variant: Find a mixed straight-and-curved perimeter
Solution
The circumference of the large circle is 160 cm, so
$2\pi R=160$
$R=(80)/(\pi)$
The three equal small circles fit along the diameter of the large circle, so
$2R=6r$
$r=(R)/(3)=(80)/(3\pi)$
Area of one small circle is
$\pi r^2 = \pi((80)/(3\pi))^2$
$=226.35\dots$
Answer: 226cm^2 to 3 significant figures.
PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

FINAL ANSWER